

PAMELA QUARTSON

Berkeley, CA | 510-717-6891 | pamela.quartson@berkeley.edu | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL PROFILE

Data engineer with experience building ELT pipelines, automated data quality checks, and a knowledge graph-based retrieval system. Comfortable working with databases (structured and unstructured), semantic data models, and owning pipeline reliability across multiple data sources.

EDUCATION

UC Berkeley — Master of Information Management and Systems, Certificate in Applied Data Science	May 2026
Ashesi University — BSc, Computer Science	May 2021

SKILLS

Languages: Python, SQL, JavaScript/TypeScript, PHP, Java, R, C++.

Data Engineering: ELT pipelines, data validation, data quality monitoring, third-party API integration

Semantic & Graph: Knowledge graphs, Memgraph, ChromaDB, ontology-aligned data extraction (LangChain)

Infrastructure: GCP, AWS, PostgreSQL, MySQL, Docker, Redis, REST APIs

WORK EXPERIENCE

Lead Developer UC Berkeley Carbon Trading Project VROD	Sep 2025 – Present
<i>VROD centralizes 10,000+ carbon offset records from six global registries; downloaded 2,100+ times in 2026 by researchers, traders, and policymakers.</i>	

- Added a new data registry end to end: integrated the third-party API and built the ELT pipeline from scratch through to production.
- Built automated data quality checks in Python that catch errors before each release, cutting manual QA time by 30%.
- Own pipeline reliability and monitoring across all six data sources and troubleshoot issues as they come up.
- Diagnosed and fixed logic flaws in an existing pipeline, then extended it to handle new transaction types.
- Lead a 3-person engineering team, running sprint planning, code reviews, and mentorship.

Web Developer UC Berkeley Goldman School of Public Policy	Jun 2025 – Sep 2025
- Co-developed an AI chatbot for the Public Policy Institute of California. - Built backend API endpoints in PHP Laravel that connect to Amazon Bedrock AI models, backed by a MySQL database for application data. - Built the React frontend, handling real-time messaging and state management.	

Software Engineer (Full-Stack) Street Streams Limited	Jul 2022 – Jul 2024
<i>Brorno is an e-commerce platform for micro-merchants in Ghana's informal trade sector.</i>	
- Designed and built the cloud-native backend from scratch on GCP as the founding engineer, including document-based and SQL data stores for the platform. - Built a real-time analytics dashboard that replaced manual daily database queries, cutting reporting time by 80% - Found that API slowness came from sequential processing with no caching. Fixed it by parallelizing with worker threads and adding Redis caching, cutting response times by 25%	

Software Engineer (Full-Stack) 518Innovations	Jun 2021 – Jun 2022
- Owned backend-heavy work for client projects: designed and built MySQL databases, then built Laravel backend logic on top of them, including the data model for an online drugstore for a pharmaceutical client. - Built and integrated PHP Laravel REST APIs, testing endpoints with Postman before release. - Used Docker for containerized deployment. Caching and query optimization improved API response times by 10%.	

PROJECTS

Privacy Regulation GraphRAG System (2025)
- Built a retrieval system comparing GraphRAG against baseline RAG for privacy regulation queries - Replaced static vector files with ChromaDB and Memgraph - Used LangChain and zero-shot prompting for knowledge graph extraction - Added an agentic workflow that updates data sources automatically - Measured a 21% accuracy gain and 17% precision gain over baseline RAG